STEPS BASED ON THE RELATED WORK ON IMAGE CLASSIFICATION :

http://venividiwiki.ee.virginia.edu/mediawiki/index.php/FPGA_Accelerated_Image_Classification_using_Neural_Networks_-_Archana_Narayanan

Image Classification using BNNs - Debugging History

1. Initial Setup

- Started a Jupyter notebook session to run the image classification project using BNNs (Binary Neural Networks).
- Imported necessary libraries such as `Xlnk`, `Overlay`, `bnn`, `numpy`, `matplotlib`, and others.
- Successfully printed the available BNN networks using `bnn.available_params(bnn.NETWORK_CNV)`.

2. First Encountered Error

- When trying to instantiate the classifier using:

```
classifier = bnn.CnvClassifier(bnn.NETWORK_CNV)
```

an error occurred:

```
OSError: cannot load library '/usr/local/lib/python3.6/dist-packages/bnn/libraries/road-signs-cnv-pynq.so': No such file or directory.
```

Image Reference:

(Insert Image: Image showing first OSError in notebook)

3. File Verification Step

- Used a script to list the contents of the target directory:

```
import os
print(os.listdir("/usr/local/lib/python3.6/dist-packages/bnn/libraries/"))
```

- Output indicated the presence of some `.so` files, but notably `road-signs-cnv-pynq.so` was missing.

Image Reference:

(Insert Image: Directory listing output)

4. Search Strategy

- Learned from documentation to try finding the missing `.so` library manually.
- Suggested command:

```
find / -name "libsdsl_lib.so" 2>/dev/null
```

(but adapted for the missing `road-signs-cnv-pynq.so` library)

5. Exploration in Jupyter Environment

- Navigated in Jupyter Notebook to `/BNN-PYNQ/bnn/libraries/`.
- Found two folders `pynqZ1-Z2` and `ultra96`, but both were empty.

Image References:

(Insert Image: Empty pynqZ1-Z2 folder)

(Insert Image: Navigation in Jupyter folders)

6. Conclusion and Next Steps

- Concluded that no precompiled `.so` files were available in the current environment.
- **Action Plan:**
  - Compile the hardware libraries manually.
  - Suggested commands:

```
cd ~/BNN-PYNQ/bnn
make clean
make hw
```

- This will build the missing `.so` files and place them properly.

Summary of Issues Encountered

- Missing shared object (`.so`) files required for BNN operation.
- Confirmed that expected `.so` files were absent from the installation directory.
- Necessary next step: Compilation of the hardware using provided `Makefile`.

Recommendations

- Always verify `.so` libraries after installation or download.
- Keep Vivado HLS ready for compilation steps.
- Prefer precompiled releases if available to save time.